

ROCmRoute Zero

Verified local intelligence on AMD ROCm — zero external inference tokens.

Correctness first. External tokens never.

The constraint

Answer diverse hidden tasks accurately while minimizing counted external model usage.

The risk

A single generic model response can violate exact formats, counts, labels, or required evidence.

Our approach

Route provable work to deterministic solvers and uncertain work to verified local inference.

The result

A local-only pipeline with explicit schema validation and no Fireworks fallback path.

Three execution layers turn prompts into verified answers

PROVE

GENERATE

VERIFY

Deterministic solvers
Arithmetic and structured tasks execute
without model generation.

Local Qwen on ROCm
vLLM serves Qwen2.5-Coder-7B entirely
on the AMD GPU.

Constraint repair
ROVL checks exact structure, content
requirements, and formatting.

The AMD notebook validated the core claim

RETIRED PUBLIC VALIDATION

A 16-task end-to-end run on ROCm 7.2, PyTorch 2.9, and vLLM 0.16.1. Public validation demonstrates readiness but does not guarantee hidden-set accuracy.

100%

16 / 16 public tasks

0

external tokens and calls

292

automated tests passing

Submission behavior is reproducible and fail-closed

Offline at runtime

The Qwen weights are bundled during image build; judging performs no model download.

Automatic startup

The entrypoint launches vLLM, waits for health, and then processes `/input/tasks.json`.

Complete output

Every task ID must appear once, in order, with a non-empty answer.

Strict egress policy

External fallback is hard-disabled and Fireworks credentials are explicitly cleared.

Local proof. Local generation. Verified output.

AMD ROCm 7.2

Qwen2.5-Coder-7B + vLLM

Zero external inference tokens